Electronic and Telecommunication Engineering

Section 2

X-ray Imaging

Rukshani Liyanaarachchi, PhD.

Acknowledgement: Dr. Nuwan Dayananda for the content

BM3122- Medical Imaging
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First X-ray

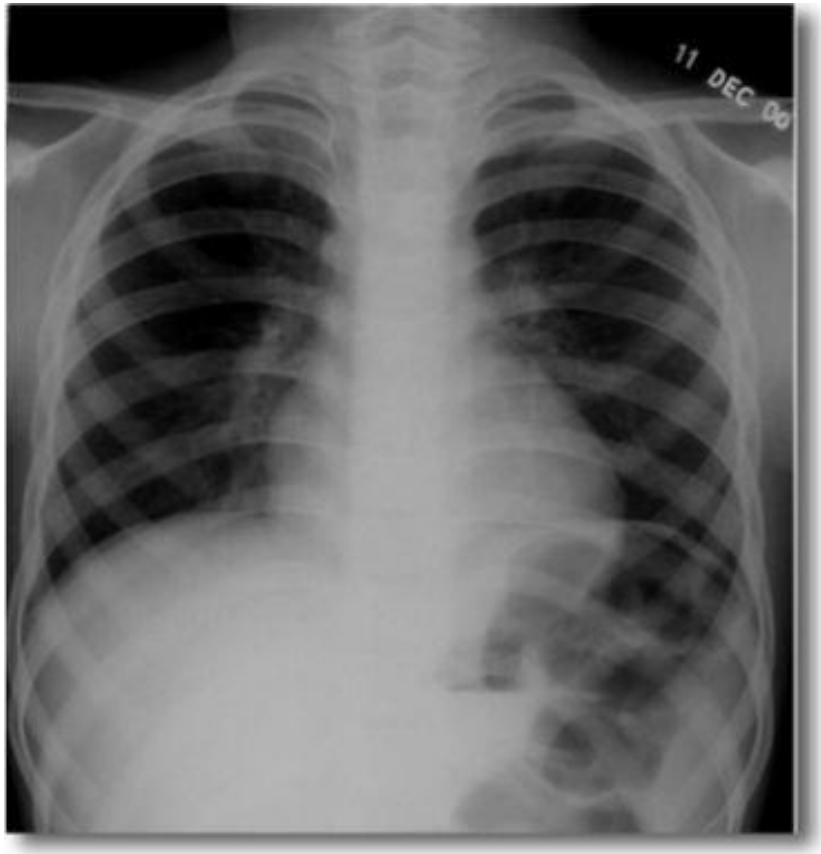
- Produced by Wilhelm Conrad Roentgen
  - His wife's hand with a ring on her finger
  - On photographic plates
  - Received the first Nobel Prize for Physics (1901)
  - Unfortunately, Roentgen, his wife, and his laboratory workers all died prematurely of cancer



Projection X-Ray (Radiography)

- Examples of modern X-ray images





Projection X-Ray (Radiography) cont.

- **2D** projections of **3D** data

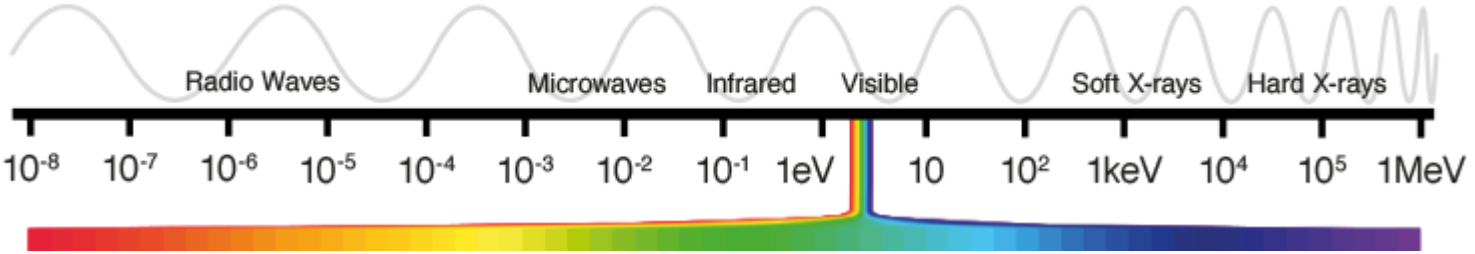
- Transmitting X-ray photons through the body onto an X-ray sensitive detector forms images
  - Some of the photons interact with tissue in the body and are **scattered or absorbed**
  - Other photons make it to the detector and are recorded

Projection X-Ray (Radiography) cont.

- The resulting image is a transmission image showing the amount of X-ray **attenuation** along a path through the body
  - Dense materials **absorb** more X-ray photons than less dense materials
  - Different absorption characteristics allow us to distinguish different materials
  - X-ray attenuation is measured by the **linear attenuation coefficient (μ)**

What is X-ray?

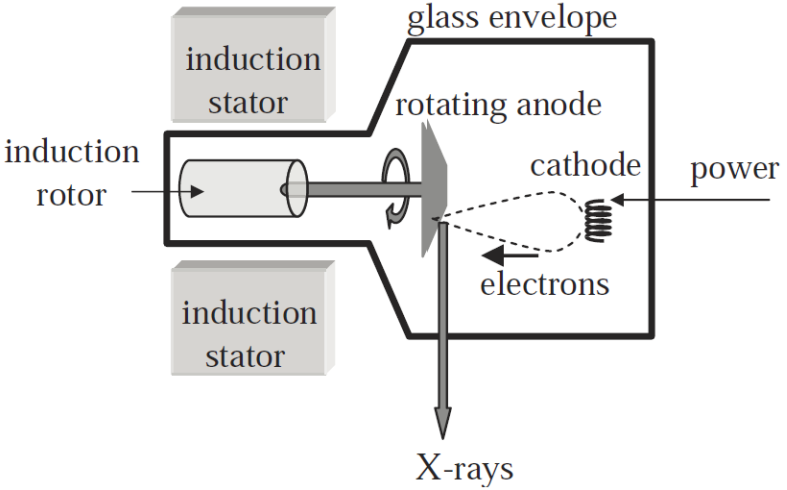
- An X-ray is **electromagnetic (EM) radiation**
- X-rays are also characterized as particles called **photons**
  - Photons with energy greater than about **2-3 eV**, are capable of ionizing atoms
  - Diagnostic X-ray radiation: **100 nm to about 0.01 nm** in wavelength (12 eV to 125 keV in energy)



X-ray Tube

- Heated filament 'boils' off electrons which are accelerated towards the anode by the applied high voltage (20 – 100 kV)
  - When electrons are decelerated at the anode, part of the energy (about 1%) is released as X-ray.
  - The rest of the energy is released as heat
  - The anode is rotated to avoid melting down a single spot and to dissipate heat out of the tube
  - All components are contained within an evacuated vessel to avoid electrons hitting any other particle.





X-ray Tube cont.



Source: Daniel Greenup for the University of Queensland

X-ray Tube cont.

- To achieve a well-defined small area in which the X-rays are created:
  - Narrow beams of electrons are produced by placing a negatively-charged focusing cup constructed around the cathode
  - The anode is bevelled at an angle between 8^0 and 17^0

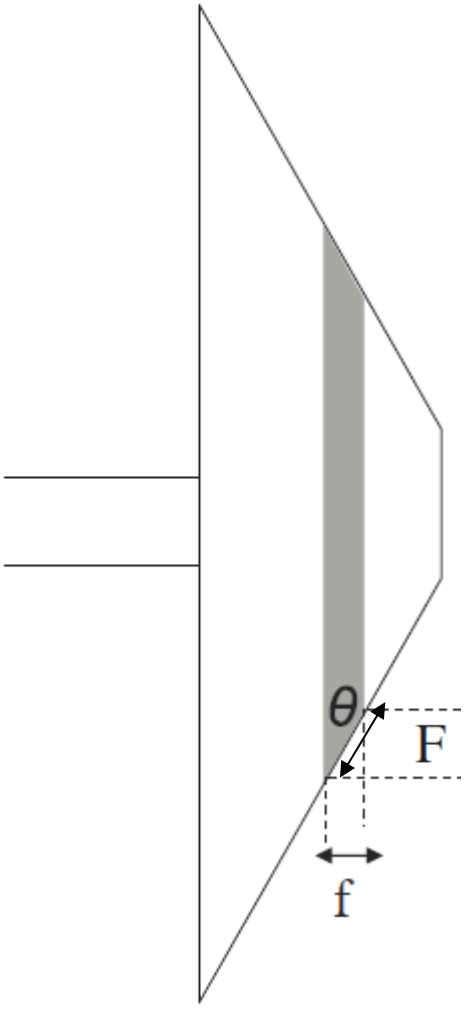
X-ray Tube cont.

- The smaller the angle, the smaller the effective focal spot size (*f*):

$$f = F \sin\theta$$

where θ is the **bevel angle** and *F* the **width** of the electron beam projection onto the anode

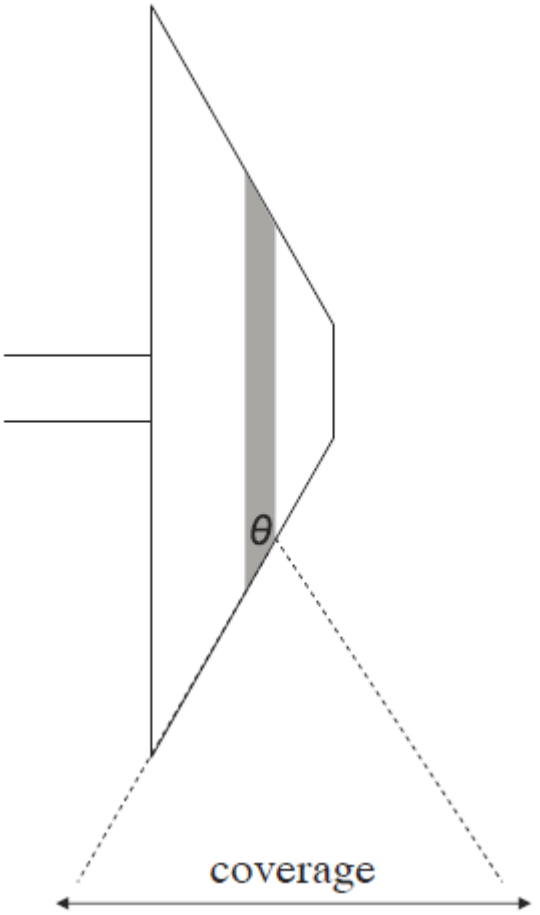
  - Example: $f \approx 0.3$ mm for digital mammography and $0.6 < f < 1.2$ mm for planar radiography and computed tomography



X-ray Tube cont.

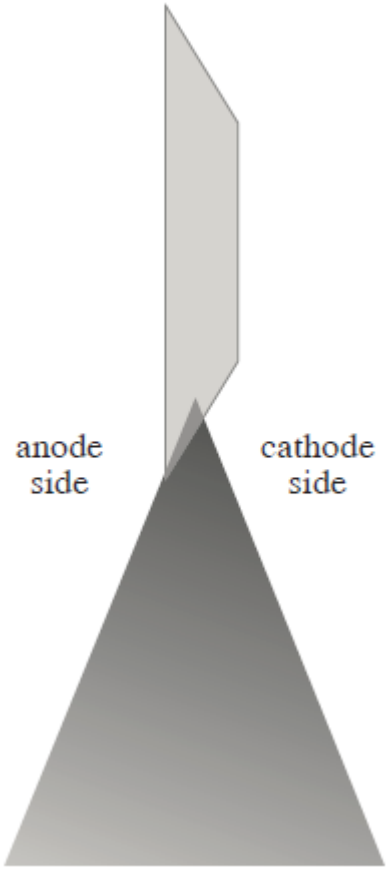
- The bevel angle also affects the coverage of the X-ray beam

$$Coverage = 2\,(source\ to\ patient\ distance) \times \tan\theta$$



X-ray Tube cont.

- Heel effect
  - In practice, the X-ray beam has a higher intensity at the ‘cathode-end’ than at the ‘anode-end’
  - The distances that X-rays have to travel is longer for X-rays produced at the anode-side of the target than at the cathode-side
  - X-rays on the anode side have to travel further through the anode itself before leaving the tube, and are therefore more attenuated
  - Can be corrected by post-processing, not significantly detrimental to the diagnostic quality of the images



X-ray Tube cont.

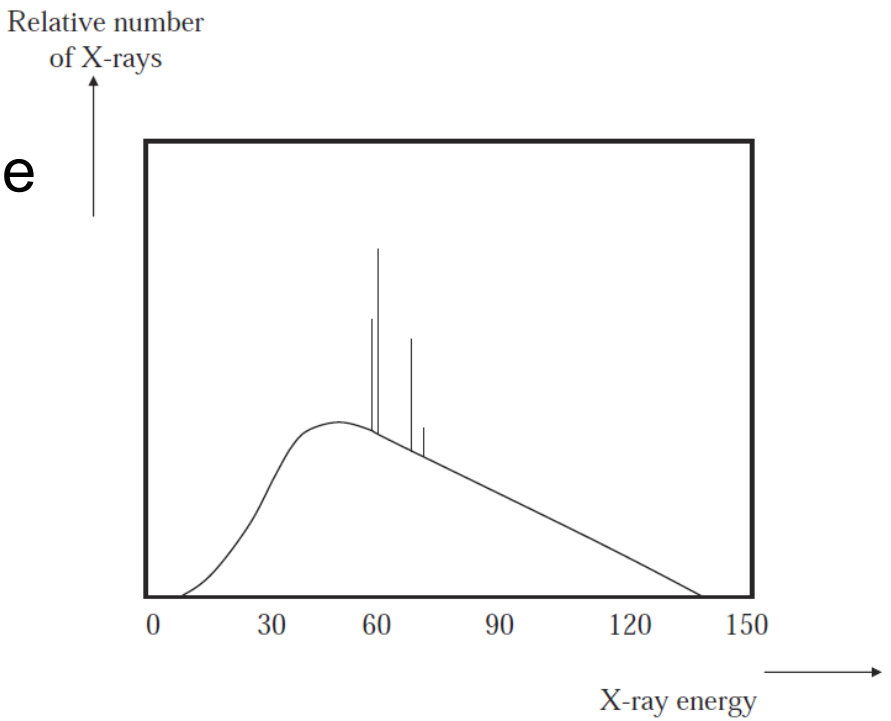
- Parameters by the operator for X-ray imaging:
  - Accelerating voltage (kV_p)
  - Tube current (mA)
  - Exposure time

- Typical values:
  - kV_p: from ~**25 kV** for digital mammography to ~**140 kV** for bone and chest applications
  - Tube current: between **50 and 400 mA** for planar radiography, and up to **1000 mA** for CT

- Power rating of the X-ray tube
  - Defined as the maximum energy dissipated in an exposure time of 0.1 s
  - Physical limitations for the values of kVp and tube current
  - Ultimately limited by **anode heating**

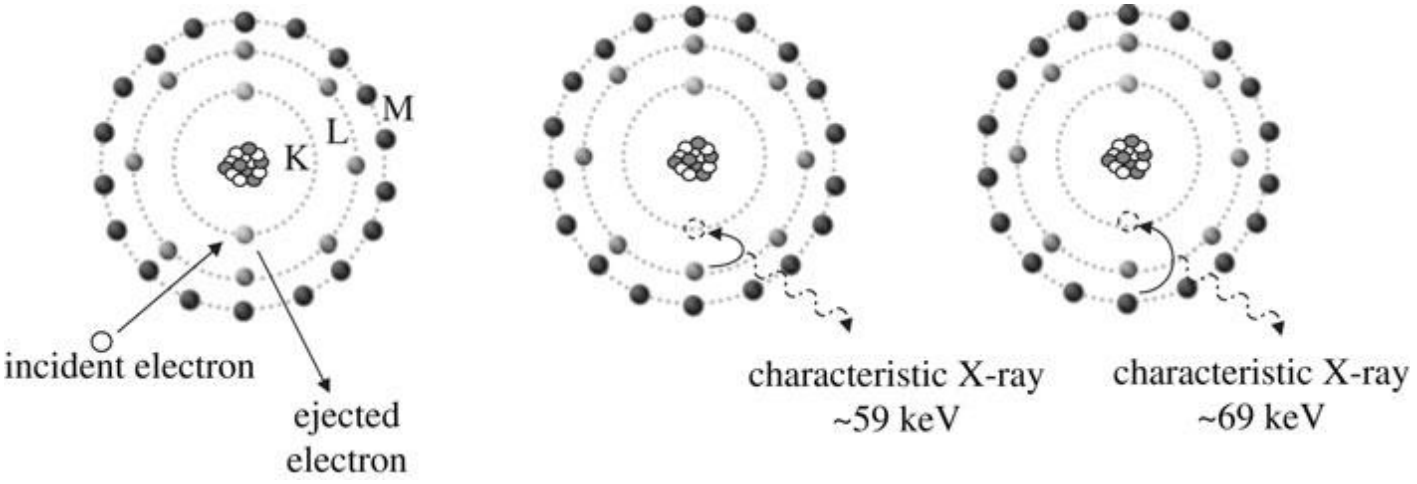
X-ray Energy Spectrum

- The plot of the relative number of X-rays produced as a function of their energy
- X-ray tubes produce X-rays with a wide range of energies
  - Up to a maximum value given by the kVp (~ **140 keV** in the spectrum below)
- Energy of the X-ray beam
  - Weighted average of all the different energies
  - Typically about two-thirds of the kVp value



X-ray Energy Spectrum cont.

- X-ray spectrum consists of sharp **characteristic** lines superimposed upon a broad energy distribution from general radiation
  - Broad energy distribution (general radiation or bremsstrahlung): due to the conversion of electron kinetic energy into X-ray energy
  - Characteristic lines (specific to the metal used for the anode): due to differences in binding energies between the ejected electrons in the inner and outer shells of atoms is emitted as a single X-ray with a specific energy



Interactions of X-rays with the body

- To produce images with high SNR and high CNR
  1. Sufficient X-rays must be **transmitted** through the body for a high SNR
  2. X-ray absorption must be sufficiently different between different tissue-types in order to produce high **contrast**
  3. X-rays which are **scattered** through unknown angles as they pass through the body must be **removed**

Interactions of X-rays with the body cont.

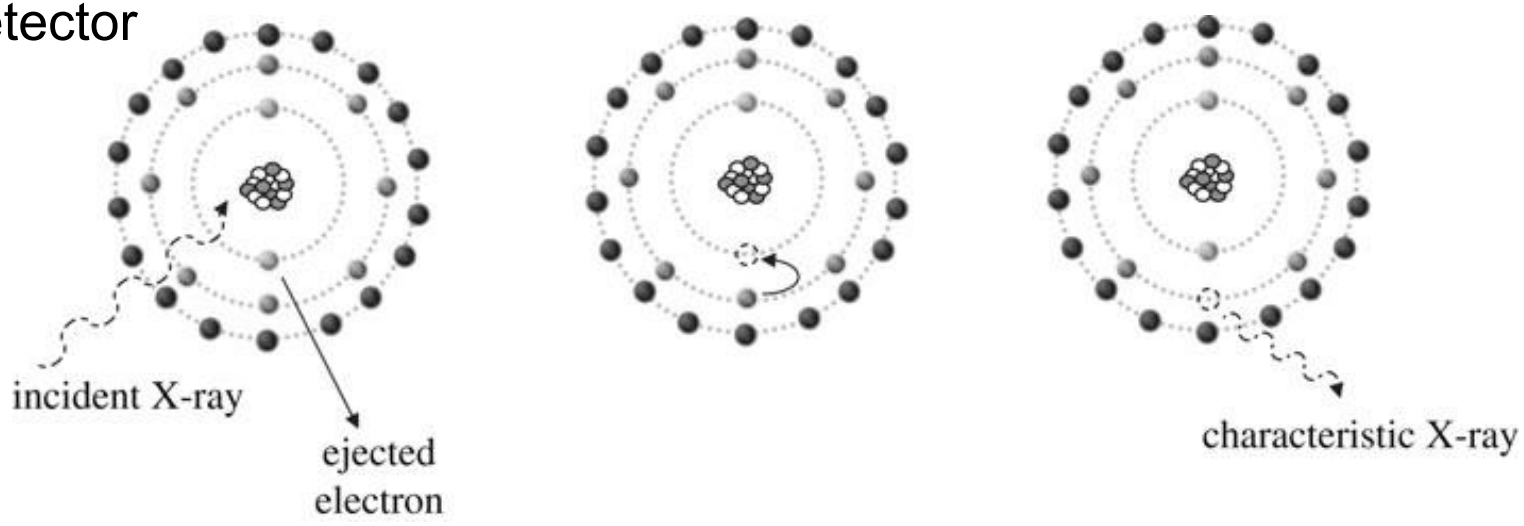
- Main mechanisms of X-ray interactions with matter
  - Coherent (Rayleigh) scattering
  - Photoelectric effect
  - Compton scattering
  - Pair production

Coherent (Rayleigh) scattering

- Coherent (Rayleigh) scattering
  - Collision of a photon with an electron such that the photon is deflected into a new direction
  - Not an ionizing interaction
  - Has a negligible effect in medical imaging
  - No change to photon energy

Photoelectric Interaction

- Tissue absorbs X-rays and provides the **contrast** in X-ray images
  - The energy of the incident X-ray is absorbed by tissue
  - A tightly bound electron being emitted from either the **K** or **L** shell
  - An electron from a higher energy level fills the 'hole' with the emission of a '**characteristic**' X-ray
  - Characteristic X-ray has a **very low energy**, a few keV at most, and is absorbed by the tissue
  - The incident X-ray is completely **absorbed** and does not reach the detector



Photoelectric Interaction cont.

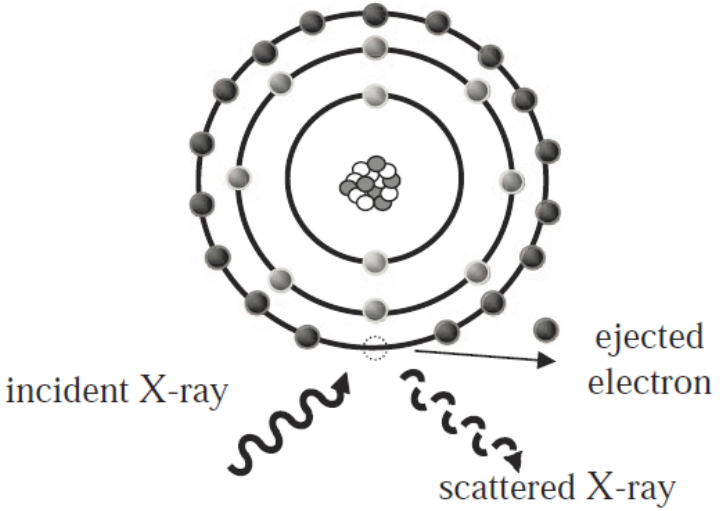
- Very important process in radiography
  - Primary or subject **contrast**
- Electron energy is **deposited** in medium
  - X-ray dose
- The probability of a photoelectric interaction (P_{PE}) occurring depends on
  - The energy (E) of the incident X-ray
  - The effective atomic number (Z_{eff}) of the tissue,
  - The tissue density (ρ)

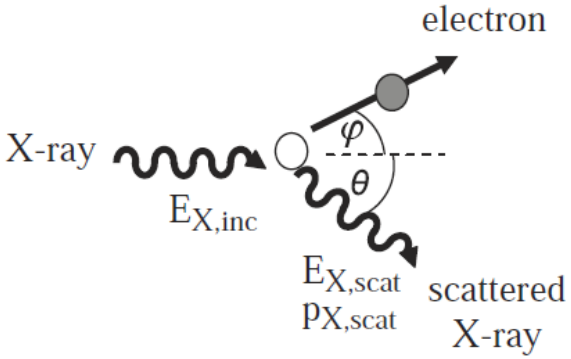
$$P_{PE} \propto \rho \frac{Z_{eff}^3}{E^3}$$

- The effective atomic number: tissue ~ 7.4, lipid ~ 6.9, and bone ~ 13.8
  - High **contrast** between bone (high attenuation) and soft tissue (low attenuation) at low X-ray energy
  - The contrast **decreases** with increasing energy

Compton Scattering

- The interaction between an incident X-ray and a loosely bound electron in an **outer** shell of an atom in tissue
  - A small fraction of the incident X-ray energy is **transferred** to this loosely-bound electron
  - The electron is **ejected** with the additional energy
  - The X-ray is deflected from its original path (by an angle θ, as shown in the figure)





Compton Scattering cont.

- The standard Compton equation to compute the change in wavelength between the incident and scattered X-ray ($\Delta\lambda$):

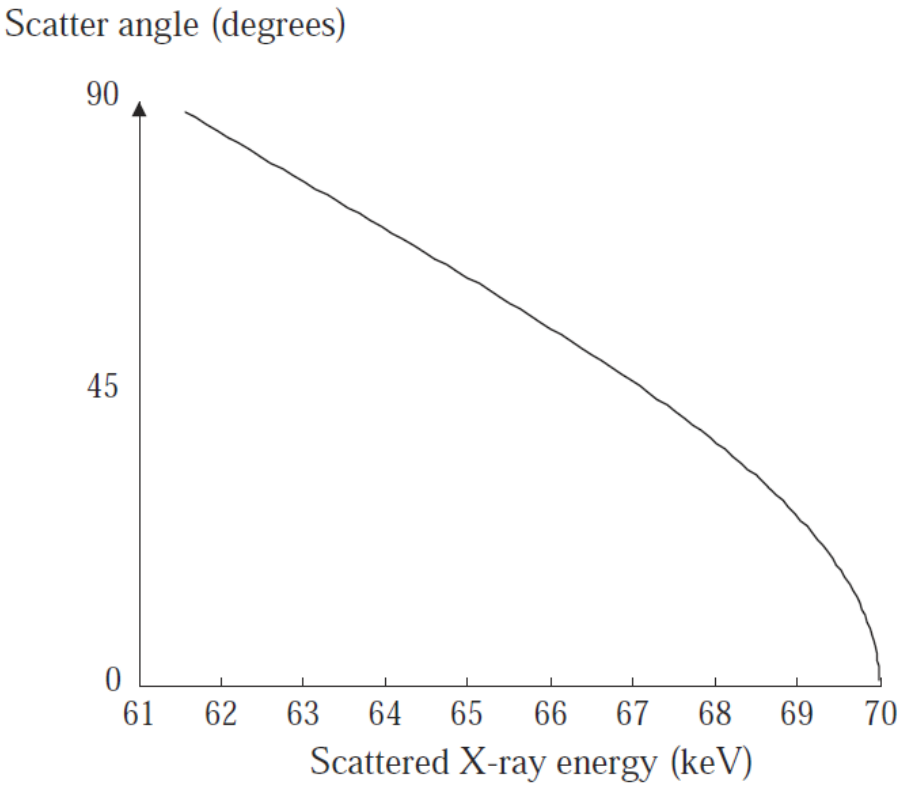
$$\Delta\lambda = \frac{h}{m_0 c}(1 - \cos\theta)$$

where h = Planck constant

m_0 = rest mass of the ejected electron

c = speed of light

Compton Scattering cont.



Example: Scatter angle as a function of the energy of a Compton-scattered X-ray for a 70 keV incident energy.

Compton Scattering cont.

- The X-ray will pass through the body and be detected as if it were primary radiation
  - X-ray energy is reduced only by a very small amount
  - The major source of noise in X-ray (and CT) images, and therefore degrades primary or subject contrast

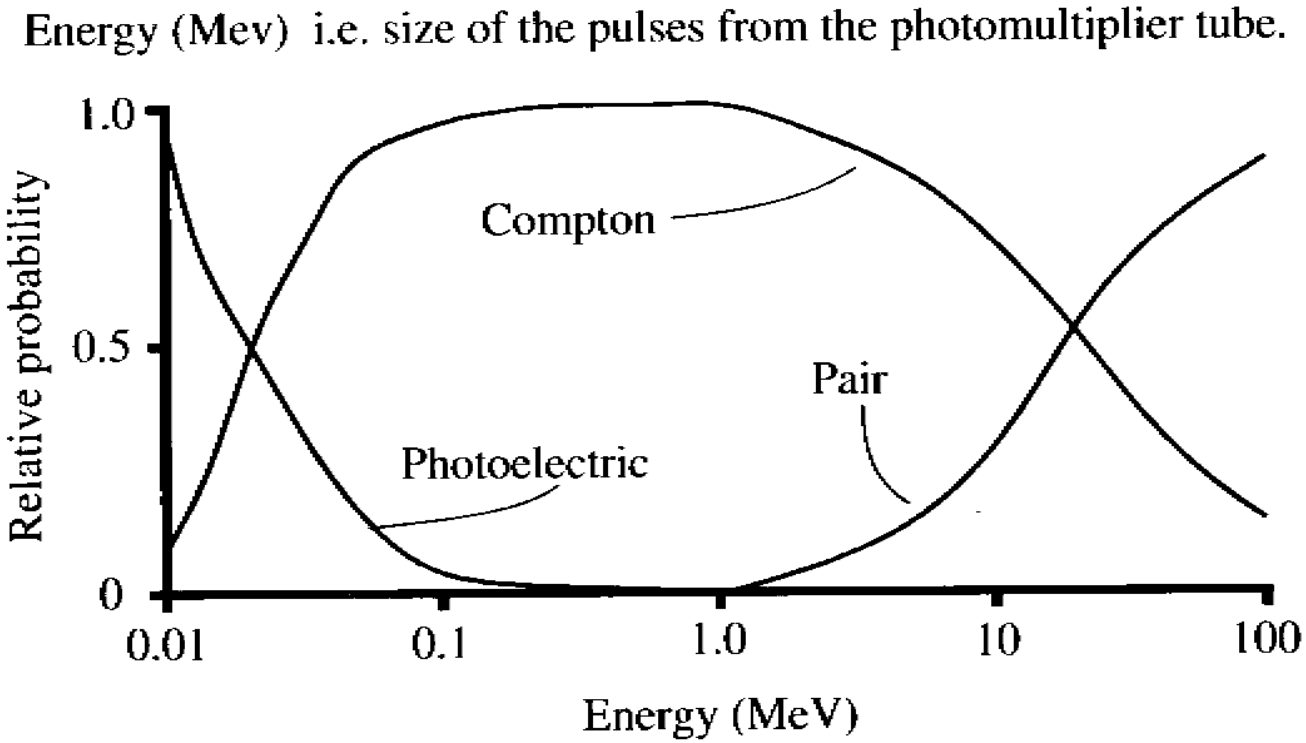
- Energy deposited in the medium
  - Dose to patient

Pair Production

- Pair production
  - High-energy photons are absorbed by a nucleus (the photon energy must be at least 1.02 MeV)
  - A positron (a positive electron: a form of anti-matter) is emitted along with an electron
  - Extremely unusual in diagnostic radiology due to the high energies required

Interaction between X-rays and matter cont.

- Relative probabilities of three absorption mechanisms



X-ray attenuation through the body

- An **exponential** process with respect to distance travelled
  - Characterized in terms of a tissue **linear attenuation coefficient**, μ
  - The value of μ depends upon the energy of the incident X-rays

- The number of X-ray photons, N, transmitted through a slab tissue of thickness, x:

$$N = N_0 \, e^{-\mu x}$$

where N_0 is of the number of incident X-rays

Linear Attenuation Coefficient

- Fraction of X-ray photons that interact per **unit thickness** of attenuator
  - Units: cm^{-1}
- Includes effects of all interactions:

$$\mu = \mu_{photoelectric} + \mu_{Compton}$$

- Depends on density of material
  - Water, ice and steam has widely different μ

Linear Attenuation Coefficient

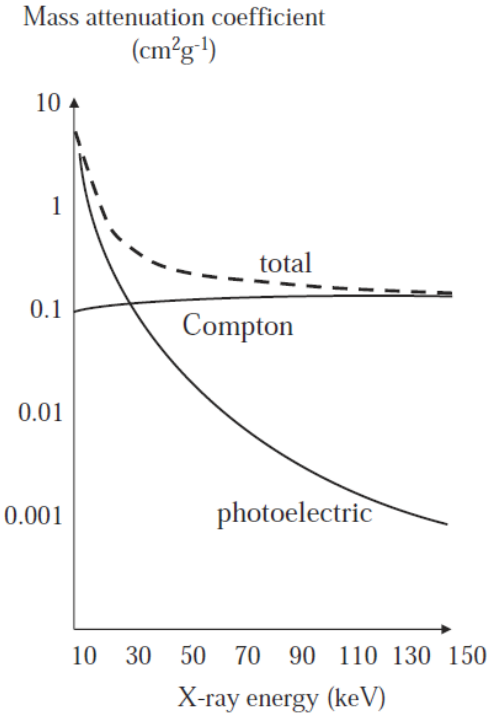
Material	Density (g/cm^3)	Electrons per Mass $(e/g) \times 10^{23}$	Electron Density $(e/cm^3) \times 10^{23}$	μ @ 50 keV (cm^{-1})
Hydrogen	0.000084	5.97	0.0005	0.000028
Water vapor	0.000598	3.34	0.002	0.000128
Air	0.00129	3.006	0.0038	0.000290
Fat	0.91	3.34	3.04	0.193
Ice	0.917	3.34	3.06	0.196
Water	1	3.34	3.34	0.214
Compact bone	1.85	3.192	5.91	0.573

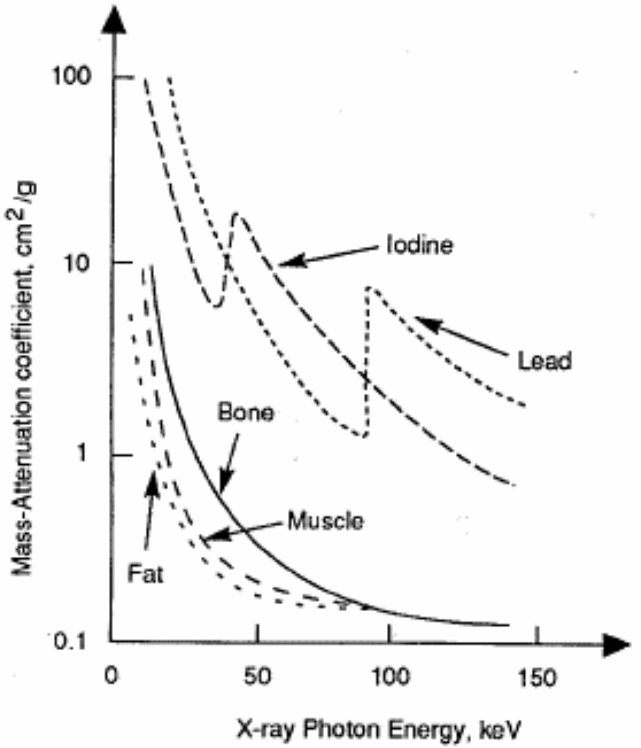
Mass Attenuation Coefficient

- Linear attenuation coefficient (μ) normalized by density (ρ)
  - Units: cm^2g^{-1}
- Mass attenuation coefficient

  Left: Contributions from different effects

  Right: Different tissue types and materials





Half Value Layer (HVL)

- The **thickness** of tissue which reduces the intensity of the X-ray beam by a factor of one-half

$$HVL = \frac{\ln 2}{\mu}$$

- Depends on the X-ray energy
  - At 100 keV: the HVL for muscle is **3.9** cm and for bone is **2.3** cm
  - At 30 keV: the HVL for muscle is **1.8** cm and for bone is **0.4** cm

Planar X-ray Radiography System

- Basic components
  1. An **X-ray tube**
  2. A **collimator** to reduce the patient dose and amount of Compton scattered X-rays
  3. An anti-scatter **grid** to reduce the effect of scattered X-rays to the image
  4. A **detector** to convert the X-ray energy into viewable format

Collimator

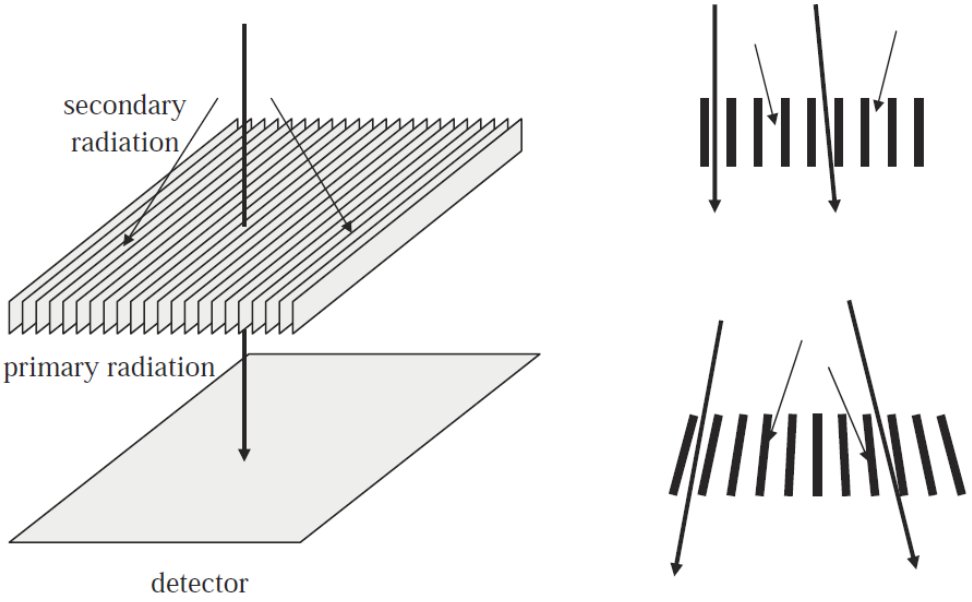
- Coverage of the X-ray beam
  - Determined by the bevel angle of the anode in the X-ray tube
  - Can be much wider than the field-of-view (FOV)
- A wider X-ray beam than the required FOV
  - Unnecessarily high patient dose
  - Increased contribution from Compton scattered X-rays from interactions with tissue outside the FOV
- A Collimator
  - Is placed between the X-ray source and the patient to restrict the dimensions of the beam within the imaging FOV
  - Consists of sheets of lead, which can be slid over one another to restrict the beam in either one or two dimensions

Grid

- Compton scattered X-rays
  - Provide little spatial information
  - Contribute to a background signal which reduces the image contrast

Grid cont.

- An anti-scatter grid
  - Is placed between the patient and the X-ray detector to reduce the contribution from scattered radiation
  - Consists of parallel strips of lead foil with aluminium used as a spacer between the strips
  - Can be manufactured at a diverging angle with a focal point ~180 cm from the grid to match the properties of the X-ray beam



Grid cont.

- Important properties
  - Thickness (t)
  - Length (h)
  - Separation (d)

- Characterized by two properties
  - Grid ratio $= \frac{h}{d}$
  - Typical values: 4:1 to 16:1
  - Grid frequency $= \frac{1}{d+t}$
  - Typical values: 5 to 7 lines/mm

- Grid can cast a shadow producing lines on images
  - Can be avoided by moving the grid forward and backward

Grid cont.

- Effects of the grid
  - Improvement in image contrast
  - Increase in the X-ray dose required to produce a specific image intensity
  - X-rays which are scattered only at very small angles (and therefore contain useful spatial information) are also absorbed
  - Characterized using a parameter called the **Bucky Factor** (BF). Typical values: 4 to 10

Bucky Factor (BF)

If an equal X-ray dose is incident upon the detector in the presence and absence of an anti-scatter grid, then the dose that is delivered to the patient increases by a factor BF.

Grid cont.

- Example: X-ray images from a model of the human pelvis

L: No anti-scatter grid, reduced CNR due to a large background signal from Compton-scattered X-rays

R: With an anti-scatter grid, reduced overall signal intensity, but improved CNR





Detector

- X-ray film
  - Traditional method of direct printing
  - Currently being replaced by digital images

- Detector technologies to produce digital images
  - Computed radiography (CR)
  - Digital radiography (DR)

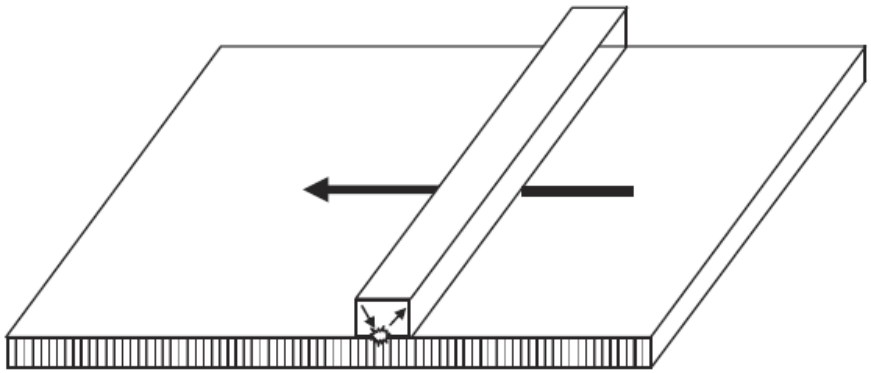
Computed Radiography (CR)

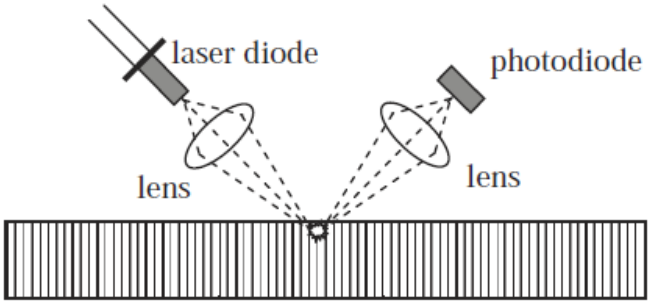
- The CR instrumentation consists of
  - A detector plate
  - A CR reader to digitize the signal on the plate

- The CR plates
  - Consist of a thin layer of phosphor crystals
  - Are protected by a light-tight cassette which is transparent to X-rays
  - Convert the X-rays which pass through the patient into light
  - Form a latent image, which is present but is not immediately visible
  - Are fed into a CR reader after the image is captured

Computed Radiography (CR) cont.

- The CR reader
  - Most common: a line-scan readout CR
  - The spatial resolution is limited by the size of the laser beam





Digital Radiography (DR)

- Indirect conversion
  - Most common method
  - X-ray energy is first converted into light by a CsI:Tl scintillator
  - Light is converted into a voltage using a two-dimensional array of photo-diodes

- Direct conversion
  - Uses direct absorption of the X-ray photons to produce an electrical signal
  - X-ray absorption is not as efficient as for CsI:Tl

Contrast Agents

- A chemical substance introduced into the body in medical imaging
  - Either **orally** or **injected** into the bloodstream
  - Enhances the contrast between that specific structure and the surrounding tissue
- Often used in X-ray, ultrasound and magnetic resonance imaging
  - Nuclear medicine techniques are entirely based on injected agents
- Designed to give the maximum **contrast** for the minimum administered **dose**
  - To have as small a degree of adverse side-effects as possible

X-ray Contrast Agents

- Very efficient in absorbing X-rays
  - Have a strong contribution from photoelectric interactions

- Basic classes of X-ray contrast agent
  1. Barium sulphate, used for gastrointestinal (GI) tract disorders and is administered orally, rectally or via a nasal cannula
  2. Agents based on water-soluble iodinated compounds which are injected into the bloodstream, and are used to visualize the vasculature in the brain, heart or peripheral arteries and veins

Contrast agents for the GI tract

- Barium sulphate
  - Used to investigate abnormalities of the GI tract such as ulcers, polyps, tumours or hernias
  - A very efficient absorber of X-rays
  - Fills the entire lumen so areas in which it is absent are visible as areas of low X-ray absorption
  - Example: an adenocarcinoma of the colon



Contrast agents for the GI tract cont.

- Patient administration
  - As a thick suspension in water using barium sulphate powder
  - Oral: for studies of the **upper GI tract**
  - Rectal: for investigations of the **lower GI tract**

- Double contrast agent
  - Barium sulphate is administered first, followed by air
  - Barium sulphate coats the inner surface of the GI tract
  - Air distends (cause to swell by pressure) the lumen
  - Used to characterize small pathologies in the large intestine, colon and rectum

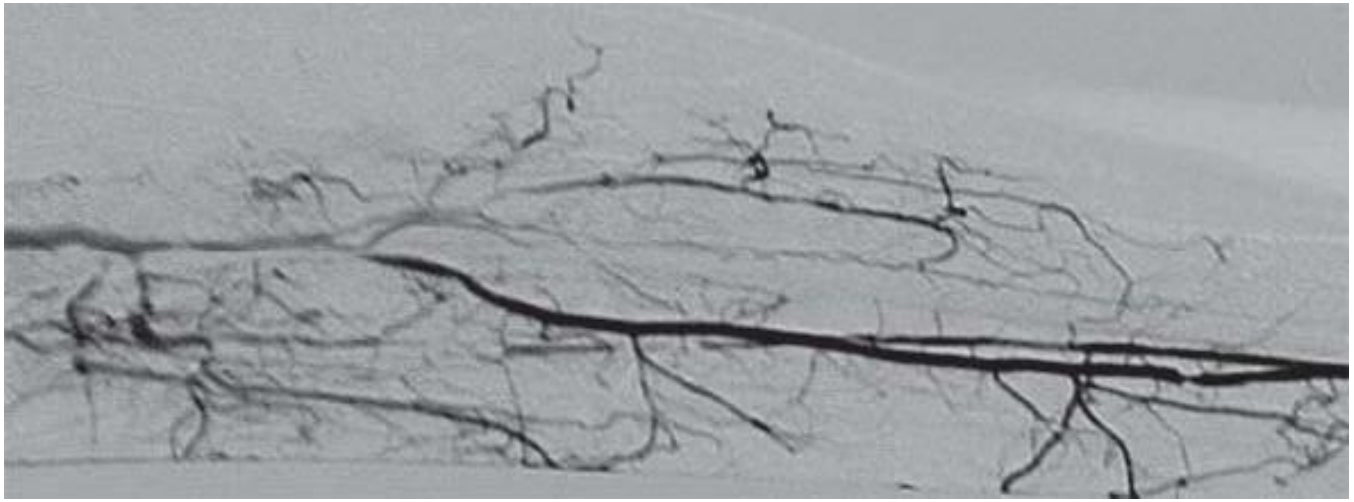
Iodine-based Contrast Agents

- Administered **intravenously** into the patient
- More than **40 million** clinical sessions are performed every year
  - One major use is **Digital Subtraction Angiography (DSA)**
  - Other applications include intravenous urography, pyelography and cholangiography
- Substantially increases X-ray attenuation in blood

Digital Subtraction Angiography (DSA)

- Used to investigate diseases such as **stenoses** and **clotting** of arteries and veins, and irregularities in systemic blood flow
- Produces very high-resolution images of the **vasculature** in the body
  - Resolves small blood vessels less than 100 μm in diameter
- Procedure:
  - Acquiring a regular image
  - Injecting a bolus of iodinated contrast agent into the bloodstream,
  - Acquiring a second image
  - Performing image **subtraction** of the two digital images

Digital Subtraction Angiography (DSA) cont.



Digital Mammography

- Used to detect small tumours or microcalcifications in the breast
  - Very high spatial resolution and CNR are needed to detect pathologies less than 1 mm in diameter
  - A low radiation dose is especially important to avoid tissue damage
- A specialized X-ray tube is used
  - Anode made from **molybdenum**, which has two K-edges at 17.9 and 19.6 keV (rather than tungsten)
  - The cathode filament is **flat**, rather than helical, in order to produce a more focused electron beam
  - Bevel angle of the anode is reduced to produce an effective focal spot size of **0.3 mm or less**
  - A beryllium window, instead of glass is used to reduce the degree of filtering of the low energy X-rays
  - A molybdenum (or aluminium) filter (30 μm thickness) is used to reduce the amount of high energy X-rays (>**20 keV**) reducing patient dose

Digital Mammography cont.

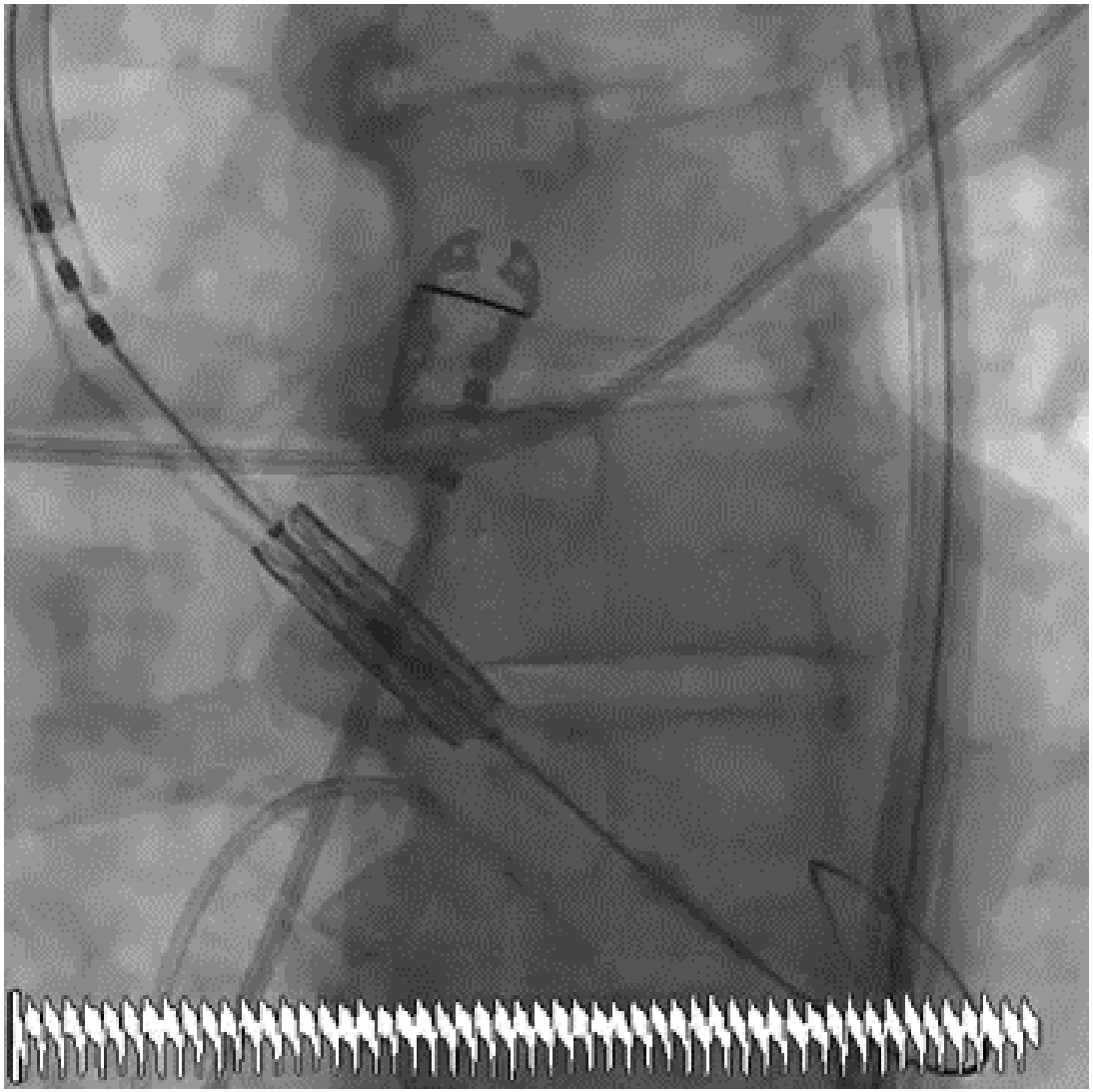
- Compression of the breast, normally to about 4 cm thickness is required to improve X-ray transmission and to reduce the contribution from Compton scatter



Digital Fluoroscopy

- Continuous X-ray imaging
  - Used to monitor **interventional** surgery
  - Example 1: Monitoring the placement of catheters, guide-wires, stents, and pacemakers
  - Example 2: Dynamic studies of the GI tract and cardiovascular system using contrast agents
- Detectors
  - Traditional: Image Intensifier (analogue or digital)
  - Modern: solid-state detectors
- Typically acquired at rates of up to 30 frames per second
- X-ray dose per frame during fluoroscopy can be as low as one-thousandth of that used during serial image acquisition

Digital Fluoroscopy cont.



Exposure and Absorbed Dose

- X-ray causes ionization primarily due to
  - Photoelectric effect
  - Compton scattering
- Absorption of energy
  - Electrons deposit energy within the vicinity of the site of their creation
  - Absorbed dose is measured in terms of **energy absorbed per unit of mass** of tissue
  - Unit of dose is the '**Gray**' (**Gy**)

  1 Gy = 1 Jkg^{-1}
  - An old unit of dose that is still used

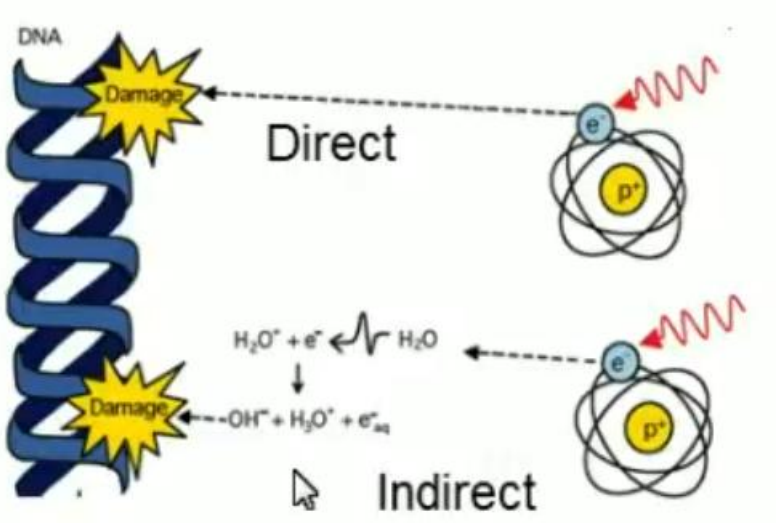
  The rad (radiation absorbed dose)

  1 rad = 0.01 Gy = 0.01 Jkg^{-1}

Radiation Dose

- Adsorbed X-ray energy in tissue from ionization

- Biological effects causing cancer induction and hereditary effects
  - Damage to DNA (breaks in DNA strands)
  - Oxygen free radicals



Equivalent Dose

- To compare biological effects of different types of radiation
  - Example: β and α particles, and proton and neutron
- Dose **normalized** to equivalent biological effect
  - Dose that gives the same risk of damage or detriment to health whatever the type of radiation
- Unit is called the Sievert (Sv)
  - 1 Sv = **1 Gy** × **QF** (Quality Factor)
  - 1 rem = **1 rad** × **QF**
- For diagnostic radiology, QF of X-ray = 1
  - 1 Sv = **1 Gy**
  - 1 rem = **1 rad**

Effective Dose (H_E)

- Equivalent whole-body dose that produce the same incidence rate of cancer and hereditary effects as the localized dose encountered in diagnostic procedures
  - In most diagnostic radiology studies, irradiation is confined to a small region (Only a particular organ or tissue will receive radiation dose)

$$Effective\ Dose\ (Sv) = \sum w_T H_T (Sv)$$

$$w_T = Tissue\ weighting\ factor$$
$$H_T = Equivalent\ dose$$

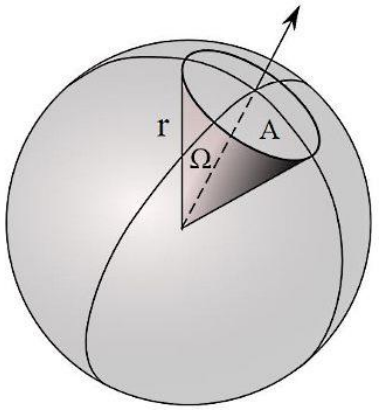
Effective Dose (H_E) cont.

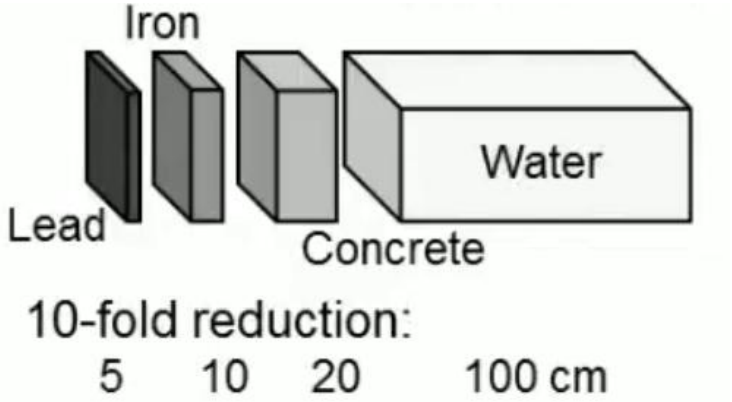
X-ray Study	$H_E\,(mSv)$
Lateral Skull	0.032
Abdomen	0.29
Chest	0.035
Lumber Spine	0.3

- Annual background dose is 2 mSv
  - Just by living
  - A frequently used comparison “yard stick”

Protection from Radiation

- Time of exposure
- Distance
  - Received radiation is proportional to the fraction of solid angle occupied by human surface A = $\frac{A}{4\pi r^2}$
- Shielding

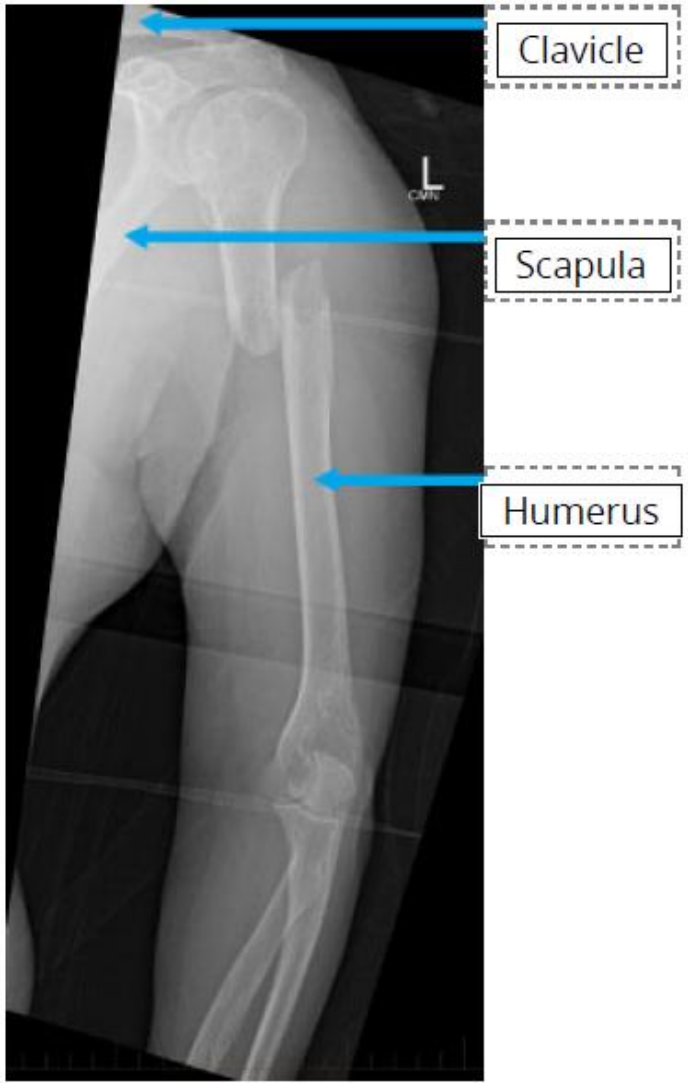




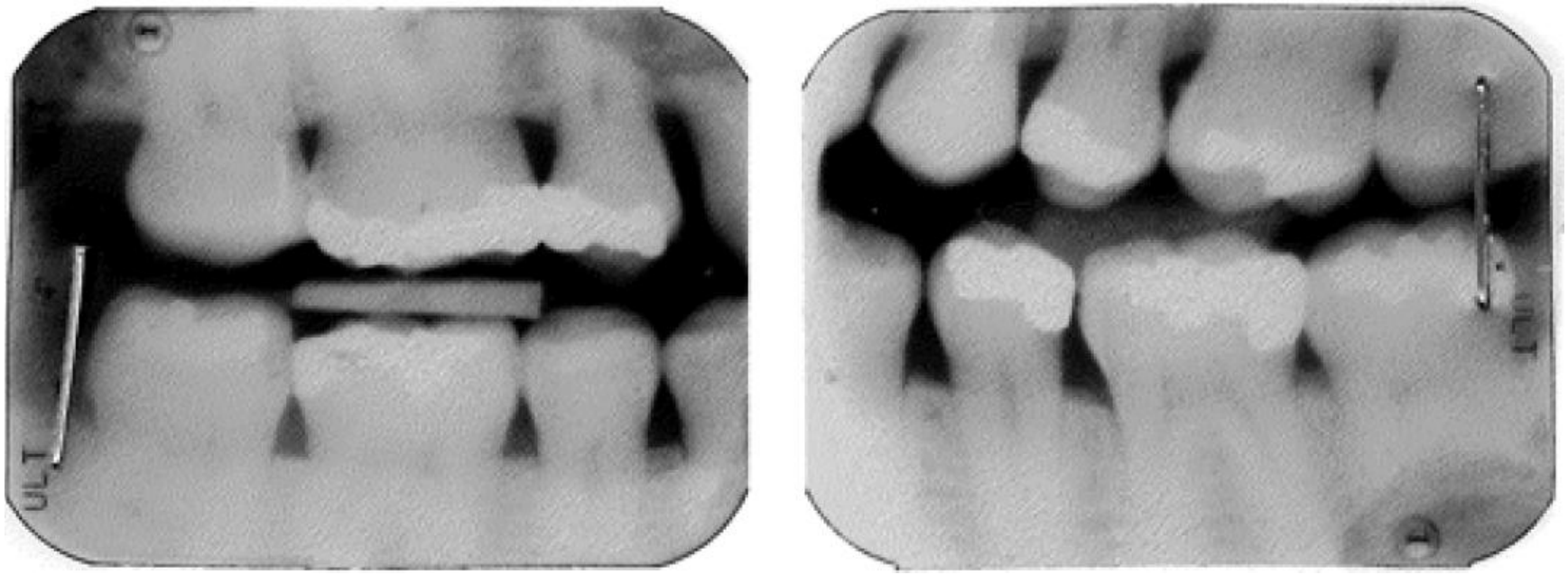
Can you identify three bones: clavicle, scapula, humerus?



Can you identify three bones: clavicle, scapula, humerus?

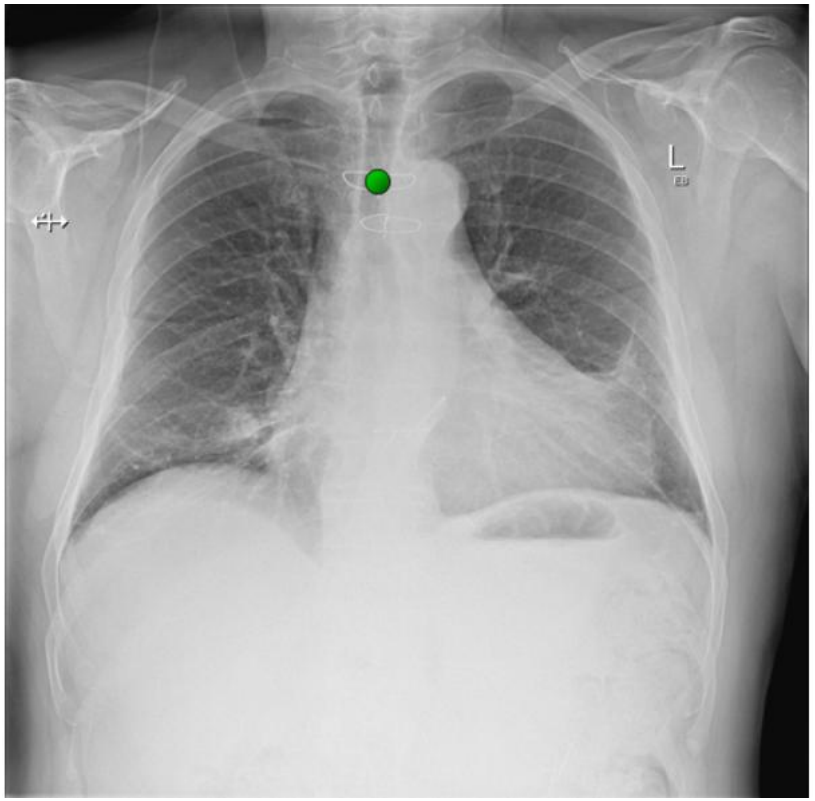


What are the hyperintense (brighter) regions?



Fillings

What are the unexpected hardware in this X-ray?



The sternal wires from an open-chest surgery.

A healthy lung is largely air, which doesn‘t absorb X-rays, and so appears dark or hypointense. In this patient, the left lung, on the right side of the image is partially consolidated, possibly due to pneumonia.

What are the darker areas circled in red on the right side of the patient?



Pockets of gas in the bowel

Remember, most X-rays are displayed so areas of high absorption are bright, or hyperintense. Dark areas are where there is very little absorption.

Intravenous Pyelogram



Here is an X-ray of the abdomen and pelvis, in which the kidneys are not visible. There is no density difference between the kidney and other organs and viscera in the abdominal cavity.

Intravenous Pyelogram cont.



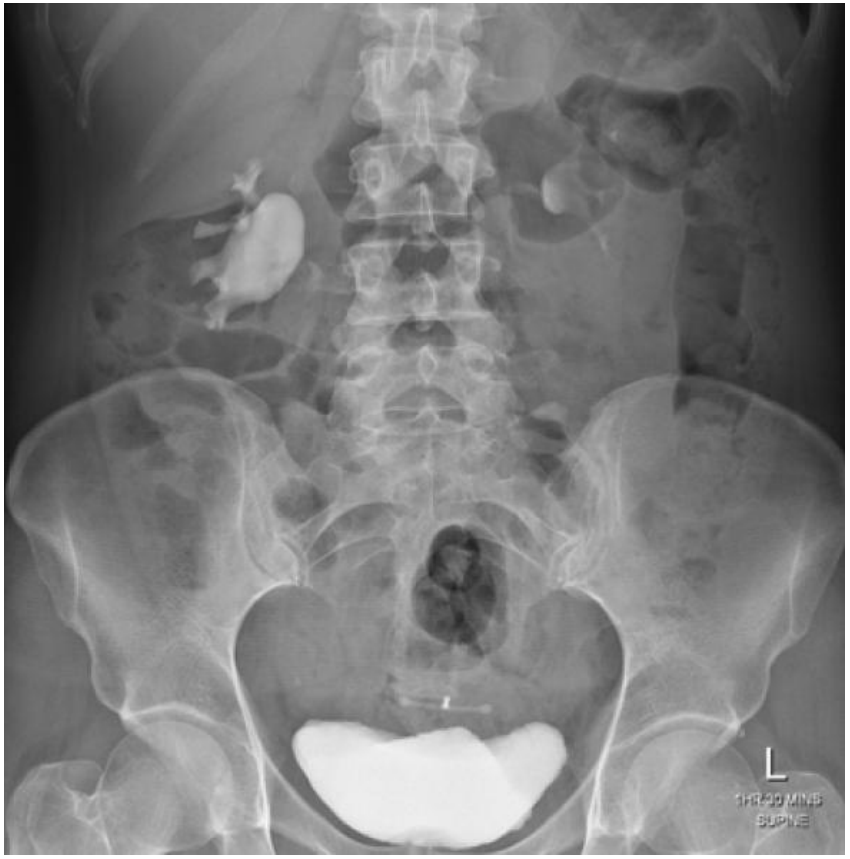
Now we can see what happens when we inject a contrast agent, which is then removed from the blood by the kidneys. The two kidneys are now visible. On the right side (the patient's left), we can see the urine in the ureter, making its way towards the bladder. In this patient, there seems to be something wrong with the right kidney. It is much larger and distended, and there doesn't appear to be any urinary flow.

Intravenous Pyelogram cont.



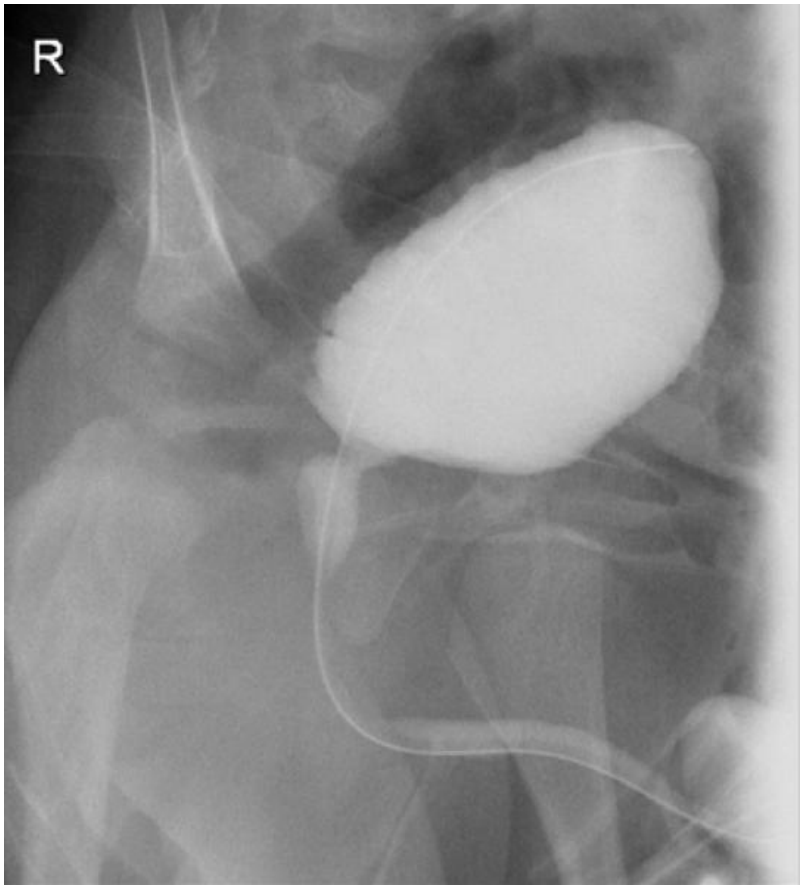
This third image confirms that the left kidney has almost emptied of dye, which has obviously drained into the bladder. The right kidney is still full of dye. The kidney is working, in that it is able to remove the waste from the blood; but it is not draining.

What is the additional hardware visible above the bladder?



An Intrauterine Device (IUD)

Micturating cysto-urethrogram



Dye is introduced into the bladder, using a catheter, and fluoroscopic imaging is employed while the patient is emptying their bladder.

Hysterosalpingogram



The contrast agent is introduced into the uterus, and then flows into the fallopian tubes. It is important in assessing potential blockages, which may lead to infertility.